

Course Code	Course Title	L	R	P	Cr
	AI-Assisted Problem Solving Using Python Programming				
Program Core					

Day 9 Session: Data Analytics with Python

Day 9

- NumPy Arrays
- Pandas DataFrames
- Matplotlib Visualization

Curiosity Question

How do AI engineers process millions of records and create meaningful visual insights using Python?

- NumPy Arrays
- Pandas DataFrames
- Matplotlib Visualization

#NumPy, Pandas, and Matplotlib are PACKAGES (libraries)

Installing Libraries

```
pip install numpy
pip install pandas
pip install matplotlib
```

Importing Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

NumPy Arrays

What is NumPy?

- Numerical Python
- Fast array processing
- Mathematical computations
- Scientific computing

Advantages of NumPy

- Faster than Lists
- Less Memory
- Vectorized Operations
- Multi-dimensional Arrays

#1. NumPy (Numerical Computing)

Key Features

- Fast array operations
- Mathematical functions

Important Functions

- numpy.array()
- numpy.zeros()
- numpy.ones()
- numpy.arange()
- numpy.mean()

#Example

```
import numpy as np
arr = np.array([1, 2, 3])
print(arr * 2)
```

Open CMD(Command Prompt)

Type: pip install numpy
to execute numpy

1. Array Creation Methods

Method	Description	Example
<code>array()</code>	Create a NumPy array	<code>np.array([1,2,3])</code>
<code>arange()</code>	Create evenly spaced values	<code>np.arange(1,10)</code>
<code>linspace()</code>	Create evenly spaced values over an interval	<code>np.linspace(0,10,5)</code>
<code>zeros()</code>	Create an array of zeros	<code>np.zeros((2,3))</code>
<code>ones()</code>	Create an array of ones	<code>np.ones((2,3))</code>
<code>empty()</code>	Create an uninitialized array	<code>np.empty((2,2))</code>
<code>full()</code>	Create an array filled with a specified value	<code>np.full((2,2),5)</code>
<code>eye()</code>	Create an identity matrix	<code>np.eye(3)</code>

2. Array Information Methods

Method	Description	Example
<code>ndim</code>	Number of dimensions	<code>a.ndim</code>
<code>shape</code>	Shape of the array	<code>a.shape</code>
<code>size</code>	Number of elements	<code>a.size</code>
<code>dtype</code>	Data type of elements	<code>a.dtype</code>
<code>itemsize</code>	Size of one element (bytes)	<code>a.itemsize</code>
<code>nbytes</code>	Total memory used	<code>a.nbytes</code>
<code>T</code>	Transpose of an array	<code>a.T</code>

3. Array Reshaping Methods

Method	Description	Example
reshape()	Change the shape of an array	a.reshape(2,3)
resize()	Resize an array	np.resize(a,(3,3))
flatten()	Convert to a 1D copy	a.flatten()
ravel()	Convert to a 1D view	a.ravel()
transpose()	Transpose an array	a.transpose()

5. Mathematical Functions

Method	Description	Example
sqrt()	Square root	np.sqrt(a)
square()	Square each element	np.square(a)
power()	Raise elements to a power	np.power(a,2)
abs()	Absolute value	np.abs(a)
exp()	Exponential	np.exp(a)
log()	Natural logarithm	np.log(a)
log10()	Base-10 logarithm	np.log10(a)

4. Arithmetic Operations

Operation	Description	Example
+	Addition	a+b
-	Subtraction	a-b
*	Element-wise multiplication	a*b
/	Division	a/b
//	Floor division	a//b
%	Modulus	a%b
**	Exponentiation	a**2

6. Statistical Methods

Method	Description	Example
sum()	Sum of elements	np.sum(a)
mean()	Average	np.mean(a)
median()	Median	np.median(a)
std()	Standard deviation	np.std(a)
var()	Variance	np.var(a)
min()	Minimum value	np.min(a)
max()	Maximum value	np.max(a)
argmin()	Index of minimum value	np.argmin(a)
argmax()	Index of maximum value	np.argmax(a)

7. Trigonometric Functions

Method	Description	Example
sin()	Sine	np.sin(a)
cos()	Cosine	np.cos(a)
tan()	Tangent	np.tan(a)
arcsin()	Inverse sine	np.arcsin(a)
arccos()	Inverse cosine	np.arccos(a)
arctan()	Inverse tangent	np.arctan(a)

8. Rounding Functions

Method	Description	Example
round()	Round values	np.round(a)
floor()	Round down	np.floor(a)
ceil()	Round up	np.ceil(a)
trunc()	Remove decimal part	np.trunc(a)

9. Searching Methods

Method	Description	Example
where()	Find elements matching a condition	np.where(a>10)
nonzero()	Indices of non-zero elements	np.nonzero(a)
argwhere()	Indices satisfying a condition	np.argwhere(a>10)

10. Sorting Methods

Method	Description	Example
sort()	Sort an array	np.sort(a)
argsort()	Return indices of sorted elements	np.argsort(a)

11. Joining Arrays

Method	Description	Example
concatenate()	Join arrays	np.concatenate((a,b))
stack()	Stack arrays	np.stack((a,b))
vstack()	Vertical stack	np.vstack((a,b))
hstack()	Horizontal stack	np.hstack((a,b))
column_stack()	Stack arrays as columns	np.column_stack((a,b))

12. Splitting Arrays

Method	Description	Example
split()	Split an array	np.split(a,2)
array_split()	Flexible split	np.array_split(a, 3)
vsplit()	Split vertically	np.vsplit(a,2)
hsplit()	Split horizontally	np.hsplit(a,2)

13. Random Module Methods

Method	Description	Example
rand()	Random floating-point values	np.random.rand(3)
randint()	Random integers	np.random.randint(1,10,5)
randn()	Normally distributed values	np.random.randn(3)
choice()	Random selection	np.random.choice(a)
shuffle()	Shuffle an array	np.random.shuffle(a)
seed()	Set random seed	np.random.seed(10)

14. Copy Methods

Method	Description	Example
copy()	Create a deep copy	a.copy()
view()	Create a shallow copy	a.view()

15. Logical Operations

Method	Description	Example
<code>all()</code>	Check if all elements are True	<code>np.all(a)</code>
<code>any()</code>	Check if any element is True	<code>np.any(a)</code>
<code>logical_and()</code>	Logical AND	<code>np.logical_and(a,b)</code>
<code>logical_or()</code>	Logical OR	<code>np.logical_or(a,b)</code>
<code>logical_not()</code>	Logical NOT	<code>np.logical_not(a)</code>

16. Aggregate Operations

Method	Description	Example
<code>cumsum()</code>	Cumulative sum	<code>np.cumsum(a)</code>
<code>cumprod()</code>	Cumulative product	<code>np.cumprod(a)</code>
<code>prod()</code>	Product of elements	<code>np.prod(a)</code>

17. Matrix Operations

Method	Description	Example
<code>dot()</code>	Dot product	<code>np.dot(a,b)</code>
<code>matmul()</code>	Matrix multiplication	<code>np.matmul(a,b)</code>
<code>transpose()</code>	Matrix transpose	<code>np.transpose(a)</code>
<code>linalg.inv()</code>	Matrix inverse	<code>np.linalg.inv(a)</code>
<code>linalg.det()</code>	Determinant	<code>np.linalg.det(a)</code>
<code>linalg.eig()</code>	Eigenvalues and eigenvectors	<code>np.linalg.eig(a)</code>

Key Features

- Data Analysis with Python
- Series
- DataFrames
- CSV Processing

What is Pandas?

- Open-source Python library
- Built on NumPy
- Used for data manipulation and analysis
- Handles structured/tabular data efficiently

Applications

- Data Analysis
- Machine Learning
- Artificial Intelligence
- Finance
- Healthcare
- Business Intelligence

#Pandas (Data Analysis) Data Structures

- Series
- DataFrame

Important Functions

- pd.read_csv()
- df.head()
- df.describe()
- df.groupby()

Open CMD(Command Prompt)

Type: pip install pandas
to execute pandas

#Example

```
import pandas as pd
data = {"name": ["Vijay", "Ram"], "age": [21, 22]}
df = pd.DataFrame(data)
print(df)
```

1. DataFrame Creation Functions

Function / Method	Description	Example
DataFrame()	Create a DataFrame	pd.DataFrame(data)
Series()	Create a Series	pd.Series([10,20,30])
read_csv()	Read CSV file	pd.read_csv("students.csv")
read_excel()	Read Excel file	pd.read_excel("students.xlsx")
read_json()	Read JSON file	pd.read_json("data.json")
read_sql()	Read data from SQL database	pd.read_sql(query, conn)

2. Data Viewing Methods

Method	Description	Example
head()	Display first rows	df.head()
tail()	Display last rows	df.tail()
sample()	Display random rows	df.sample(5)
info()	Dataset information	df.info()
describe()	Statistical summary	df.describe()

3. DataFrame Attributes

Attribute	Description	Example
shape	Number of rows and columns	df.shape
size	Total number of elements	df.size
columns	Column names	df.columns
index	Row labels	df.index
dtypes	Data types of columns	df.dtypes
values	Data as NumPy array	df.values

4. Selection Methods

Method	Description	Example
loc[]	Select rows/columns by label	df.loc[0]
iloc[]	Select rows/columns by position	df.iloc[0]
at[]	Access a single value by label	df.at[0,"Name"]
iat[]	Access a single value by index	df.iat[0,1]

6. Sorting Methods

Method	Description	Example
sort_values()	Sort by column values	df.sort_values("Marks")
sort_index()	Sort by index	df.sort_index()

5. Data Manipulation Methods

Method	Description	Example
insert()	Insert a column	df.insert(2,"Age",[20,21])
drop()	Remove rows/columns	df.drop("Age",axis=1)
rename()	Rename columns	df.rename(columns={"Name":"Student"})
replace()	Replace values	df.replace("M","Male")
assign()	Add new column	df.assign(Total=100)

7. Filtering Methods

Method	Description	Example
query()	Filter using conditions	df.query("Marks>80")
isin()	Check membership	df["City"].isin(["Delhi"])
between()	Check range	df["Marks"].between(60,90)

8. Missing Value Methods

Method	Description	Example
isnull()	Identify missing values	df.isnull()
notnull()	Identify non-missing values	df.notnull()
fillna()	Fill missing values	df.fillna(0)
dropna()	Remove missing values	df.dropna()

10. Grouping and Aggregation Methods

Method	Description	Example
groupby()	Group data	df.groupby("Dept")
agg()	Apply aggregate functions	df.agg(["sum","mean"])
pivot_table()	Create pivot table	pd.pivot_table(df)

9. Statistical Methods

Method	Description	Example
sum()	Sum of values	df["Marks"].sum()
mean()	Average	df["Marks"].mean()
median()	Median	df["Marks"].median()
mode()	Most frequent value	df["Marks"].mode()
min()	Minimum value	df["Marks"].min()
max()	Maximum value	df["Marks"].max()
count()	Count non-null values	df.count()
std()	Standard deviation	df.std()
var()	Variance	df.var()

11. Combining Data

Method	Description	Example
concat()	Concatenate DataFrames	pd.concat([df1,df2])
merge()	Merge DataFrames	pd.merge(df1,df2)
join()	Join DataFrames	df1.join(df2)

12. File Handling Methods

Method	Description	Example
to_csv()	Write to CSV	df.to_csv("data.csv")
to_excel()	Write to Excel	df.to_excel("data.xlsx")
to_json()	Write to JSON	df.to_json("data.json")
to_sql()	Write to SQL database	df.to_sql("table", conn)

14. Index Methods

Method	Description	Example
set_index()	Set a column as index	df.set_index("ID")
reset_index()	Reset index	df.reset_index()

16. Date and Time Methods

Method	Description	Example
to_datetime()	Convert to datetime	pd.to_datetime(df["Date"])
dt.year	Extract year	df["Date"].dt.year
dt.month	Extract month	df["Date"].dt.month
dt.day	Extract day	df["Date"].dt.day

13. Duplicate Handling Methods

Method	Description	Example
duplicated()	Identify duplicate rows	df.duplicated()
drop_duplicates()	Remove duplicate rows	df.drop_duplicates()

15. String Methods

Method	Description	Example
str.upper()	Convert to uppercase	df["Name"].str.upper()
str.lower()	Convert to lowercase	df["Name"].str.lower()
str.contains()	Check substring	df["Name"].str.contains("A")
str.replace()	Replace text	df["Name"].str.replace("A","X")

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to_csv()	Write to CSV	df.to_csv("data.csv")
to_excel()	Write to Excel	df.to_excel("data.xlsx")
to_json()	Write to JSON	df.to_json("data.json")
to_sql()	Write to SQL database	df.to_sql("table", conn)

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Method	Description	Example
set_index()	Set a column as index	df.set_index("ID")
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Matplotlib Visualization Python Data Visualization

- Line Charts
- Bar Charts
- Pie Charts
- Scatter Plots
- Histograms

What is Matplotlib?

Definition

Matplotlib is an open-source Python library used for creating static, animated, and interactive visualizations.

Advantages

- Easy to use
- High-quality graphs
- Supports multiple chart types
- Integrates with NumPy and Pandas
- Widely used in Data Science and AI

#Matplotlib (Visualization)

Types of Graphs

- Line chart
- Bar chart
- Pie chart
- Histogram

Open CMD(Command Prompt)

Type: pip install matplotlib
to execute matplotlib

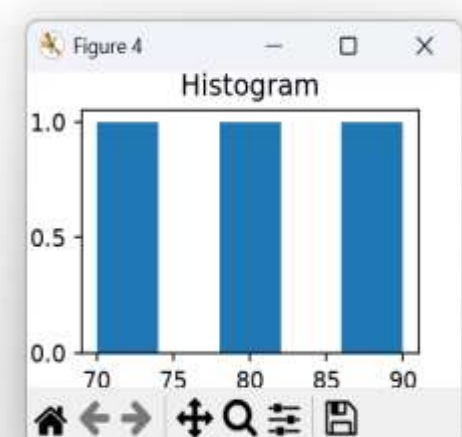
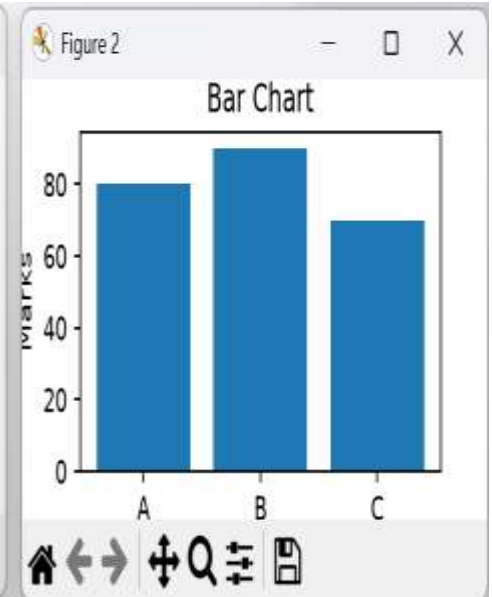
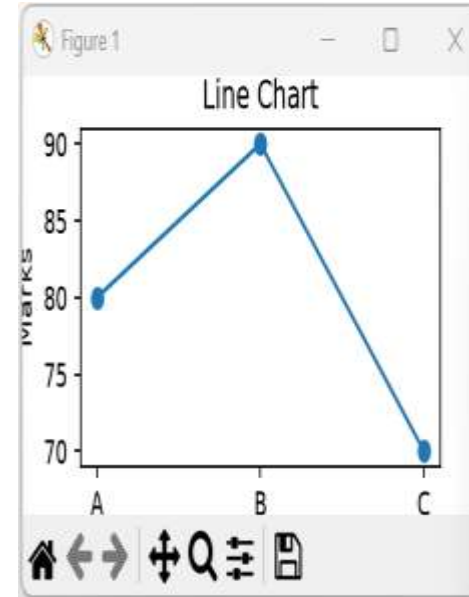
#Example 1:

```
import matplotlib.pyplot as plt
x = [1, 2, 3]
y = [10, 20, 30]
plt.plot(x, y)
plt.show()
```

#Example 2:

```
import matplotlib.pyplot as plt
names = ["A", "B", "C"]
marks = [80, 90, 70]
plt.bar(names, marks)
plt.title("Bar Chart")
plt.show()
```

```
#Example Program for all Charts
import matplotlib.pyplot as plt
names = ["A", "B", "C"]
marks = [80, 90, 70]
# ----- Line Chart -----
plt.figure(1)
plt.plot(names, marks, marker='o')
plt.title("Line Chart")
plt.xlabel("Students")
plt.ylabel("Marks")
# ----- Bar Chart -----
plt.figure(2)
plt.bar(names, marks)
plt.title("Bar Chart")
plt.xlabel("Students")
plt.ylabel("Marks")
# ----- Pie Chart -----
plt.figure(3)
plt.pie(marks, labels=names, autopct='%1.1f%%')
plt.title("Pie Chart")
# ----- Histogram -----
plt.figure(4)
plt.hist(marks, bins=5)
plt.title("Histogram")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```



Function	Description	Example
<code>plot()</code>	Draw a line graph	<code>plt.plot(x, y)</code>
<code>bar()</code>	Create a vertical bar chart	<code>plt.bar(names, marks)</code>
<code>barh()</code>	Create a horizontal bar chart	<code>plt.barh(names, marks)</code>
<code>pie()</code>	Create a pie chart	<code>plt.pie(values, labels=labels)</code>
<code>scatter()</code>	Create a scatter plot	<code>plt.scatter(x, y)</code>
<code>hist()</code>	Create a histogram	<code>plt.hist(data)</code>
<code>boxplot()</code>	Create a box plot	<code>plt.boxplot(data)</code>
<code>fill_between()</code>	Create an area chart	<code>plt.fill_between(x, y)</code>
<code>stem()</code>	Create a stem plot	<code>plt.stem(x, y)</code>
<code>step()</code>	Create a step chart	<code>plt.step(x, y)</code>
<code>xlabel()</code>	Set the X-axis label	<code>plt.xlabel("Students")</code>
<code>ylabel()</code>	Set the Y-axis label	<code>plt.ylabel("Marks")</code>
<code>title()</code>	Set the chart title	<code>plt.title("Student Performance")</code>
<code>suptitle()</code>	Set a title for all subplots	<code>plt.suptitle("Sales Dashboard")</code>

Function	Description	Example
<code>legend()</code>	Display the legend	<code>plt.legend(["Marks"])</code>
<code>grid()</code>	Show or hide the grid	<code>plt.grid(True)</code>
<code>xlim()</code>	Set X-axis limits	<code>plt.xlim(0, 10)</code>
<code>ylim()</code>	Set Y-axis limits	<code>plt.ylim(0, 100)</code>
<code>xticks()</code>	Set X-axis tick positions/labels	<code>plt.xticks([1,2,3])</code>
<code>yticks()</code>	Set Y-axis tick positions/labels	<code>plt.yticks([0,50,100])</code>
<code>figure()</code>	Create a new figure	<code>plt.figure(figsize=(6,4))</code>
<code>subplot()</code>	Create multiple plots in one figure	<code>plt.subplot(2,1,1)</code>
<code>subplots()</code>	Create a figure with multiple axes	<code>fig, ax = plt.subplots(2,2)</code>
<code>tight_layout()</code>	Adjust subplot spacing	<code>plt.tight_layout()</code>
<code>savefig()</code>	Save the chart as an image	<code>plt.savefig("graph.png")</code>
<code>show()</code>	Display the chart	<code>plt.show()</code>
<code>close()</code>	Close the current figure	<code>plt.close()</code>
<code>cla()</code>	Clear the current axes	<code>plt.cla()</code>

Function	Description	Example
<code>clf()</code>	Clear the current figure	<code>plt.clf()</code>
<code>text()</code>	Add text at a specific position	<code>plt.text(2, 50, "Topper")</code>
<code>annotate()</code>	Add annotations with arrows	<code>plt.annotate("Peak", (3,80))</code>
<code>axhline()</code>	Draw a horizontal line	<code>plt.axhline(y=50)</code>
<code>axvline()</code>	Draw a vertical line	<code>plt.axvline(x=5)</code>
<code>axis()</code>	Set or get axis limits	<code>plt.axis([0,10,0,100])</code>
<code>style.use()</code>	Apply a plotting style	<code>plt.style.use("ggplot")</code>
<code>imshow()</code>	Display an image	<code>plt.imshow(image)</code>
<code>colorbar()</code>	Add a color bar	<code>plt.colorbar()</code>



Thank you